EMPLOYEE MANAGEMENT SYSTEM(EMS)

CSE 100: Software Development Project **I**

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Abstract

The “Employee Management System (EMS)” project is a console-based software developed in C++ to manage employee-related tasks in an organization. It provides easy access to employee data and automates key processes, improving efficiency. The system offers a simple interface with two user roles: Administrator and Guest, each with different access levels. Key features of the EMS include searching and viewing employee data, saving and loading files for easy data management, and a salary grading system based on predefined criteria. It also allows storing and retrieving detailed employee information, helping HR managers and administrators track employee performance, roles, and compensation. The system provides customizable viewing options for better presentation of data. The advantages of the EMS include better organization and accuracy in managing employee data, simplified administrative tasks, and increased security through user roles. It reduces manual work and errors, improving overall productivity and ensuring that employee information is always accessible and up-to-date, aiding decision-making and efficient management.

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Chapter 1
INTRODUCTION

1.1. Problem Specification

Manual handling of employee information presents several challenges, such as slow processes in leave management, where approval can take weeks. Paper-based systems are prone to human error, loss, and unauthorized access, while being time-consuming. Many current systems lack employee self-service, requiring staff to go through an administrator to access their own data. Additionally, multinational companies face difficulties accessing employee information stored at headquarters, especially in remote locations. This project aims to create a secure, password-protected employee information system to store details such as personal information and salary, enabling efficient management and easy access to important data.[1]

1.2. Objectives

The main objectives of the Employee Management System(EMS) are:-

- To develop a user-friendly, console-based application to manage employee information.
- To automate tasks like storing, retrieving, and updating employee data, reducing manual errors.
- To implement a salary grading system based on predefined criteria, improving payroll management.
- To provide different access levels (Administrator and Guest) to ensure secure and appropriate access to employee data.
- To keep employee information up-to-date and easy to access for decision-making.

1.3. Scope

- Project Goals: Automate employee data management, implement user roles, create a salary management system, and store personal information (name, contact, address, etc.) of employees.
- Deliverables: Employee data storage, salary grading system, file handling, and user authentication.
- Tasks: System design, development in C++, testing, and documentation.
- Costs: Minimal, primarily development time.
- Deadlines: 1 week for planning, 3 weeks for development, 4–6 weeks for full completion.

1.4. Organization Of Project Report

The various chapters that make up this report's structure each focus on important project aspects:

● **Chapter 1: Introduction:** Outlines the issue, the reason behind it, its importance, its goals, and how it is constructed.

● **Chapter 2: Background:** Examines current employee management systems and techniques.
● **Chapter 3: System Analysis & Design** explains the suggested system's architecture, design tenets, and implementation specifics.
● **Chapter 4: Implementation:** Offers a thorough explanation of the development process, covering functionality and code structure.
In Chapter 5, **"User Manual,"** the admin and user user interfaces are described and shown, along with sample login credentials.
● **Chapter 6: Conclusion and Future Work**:outlines the accomplishments of the project and suggests improvements for the future.

Chapter 2
BACKGROUND

2.1. Existing System Analysis

2.1.1:

Employee Management System(manual): In a manual system, employee information, such as personal details, attendance, and payroll, is recorded and managed by hand in physical records (files, spreadsheets, paper-based forms). This could involve using notebooks, spreadsheets, or printed documents.

pros:

Low Initial Cost : Requires no specialized equipment or programs.
Simple to Use: Managing records doesn't demand technical expertise.
Customization: Easily modified without coding or software skills..

cons:

prone to errors: Record loss is a possibility, and data entering errors can occur.
Time-consuming: It takes a while to generate reports or conduct information searches.
Lack of Scalability: Manual data management is harder and more complicated as the workforce size increases.
Analyzing data is difficult because without digital tools, reporting and analysis are less effective.

2.1.2:

School Management System (Digital) : A school management system (SMS) is a software solution designed to streamline various administrative tasks in educational institutions. Here are the pros and cons of implementing such a system:

pros:

Efficient administration: Automates administrative tasks, reduces paperwork and saves time.
Enhance communication**:** Promotes transparency in the education system by improving communication between teachers, students and parents.

cons:

Implementation Costs: Initial costs for software and training can be a significant investment for schools.

Resistance to Change: Staff or parents may resist adopting new technologies, impacting the system's effectiveness.

Technical Issues: Possible technical glitches or downtime can disrupt operations if not addressed promptly. [2]

2.2. Research Gap:

The gaps in the existing system include:

1. User-Friendliness: Manual EMS and school management systems often lack intuitive interfaces. EMS should focus on simplicity and customization for different user roles to reduce training time.

2. Security and Data Privacy: Data security is sometimes overlooked by school management systems. To safeguard sensitive employee data, the Employee Management System must place a high priority on role-based access, robust encryption, and frequent audits.

3. Scalability: Scalability issues may arise with systems such as school management. EMS should be built to expand with the company, accommodating more staff members and positions.

4. Time-Consuming and Error-Prone: Manual data entry increases the chances of errors and slows down the booking process. EMS is a digitized system so it does not have these problems.

Chapter 3
SYSTEM ANALYSIS & DESIGN

3.1. Technology & Tools

The technology and tools utilized in the project are briefly described here:

Programming Language :

The project's primary programming language is C++. Because of its performance, flexibility, and support for object-oriented programming, C++ was selected.

Environment for Integrated Development (IDE) :

CodeBlocks: A well-known open-source integrated development environment for writing, compiling, and debugging C++ code. Multiple compilers are supported by Code::Blocks, which also offers an easy-to-use interface for productive development and debugging.

The compiler :

CodeBlocks' C++ code is compiled using the GNU GCC compiler. Many C++ features and optimizations are supported by this open-source compiler.

Hardware :

HP 250 G10 Notebook PC, 15.6-inch

This laptop, which has a dependable processor and a 15.6-inch display, offers a steady environment for C++ programming.

It has enough RAM and [SSD/HDD storage] to enable seamless testing, compilation, and coding.

3.2. Model & Diagram

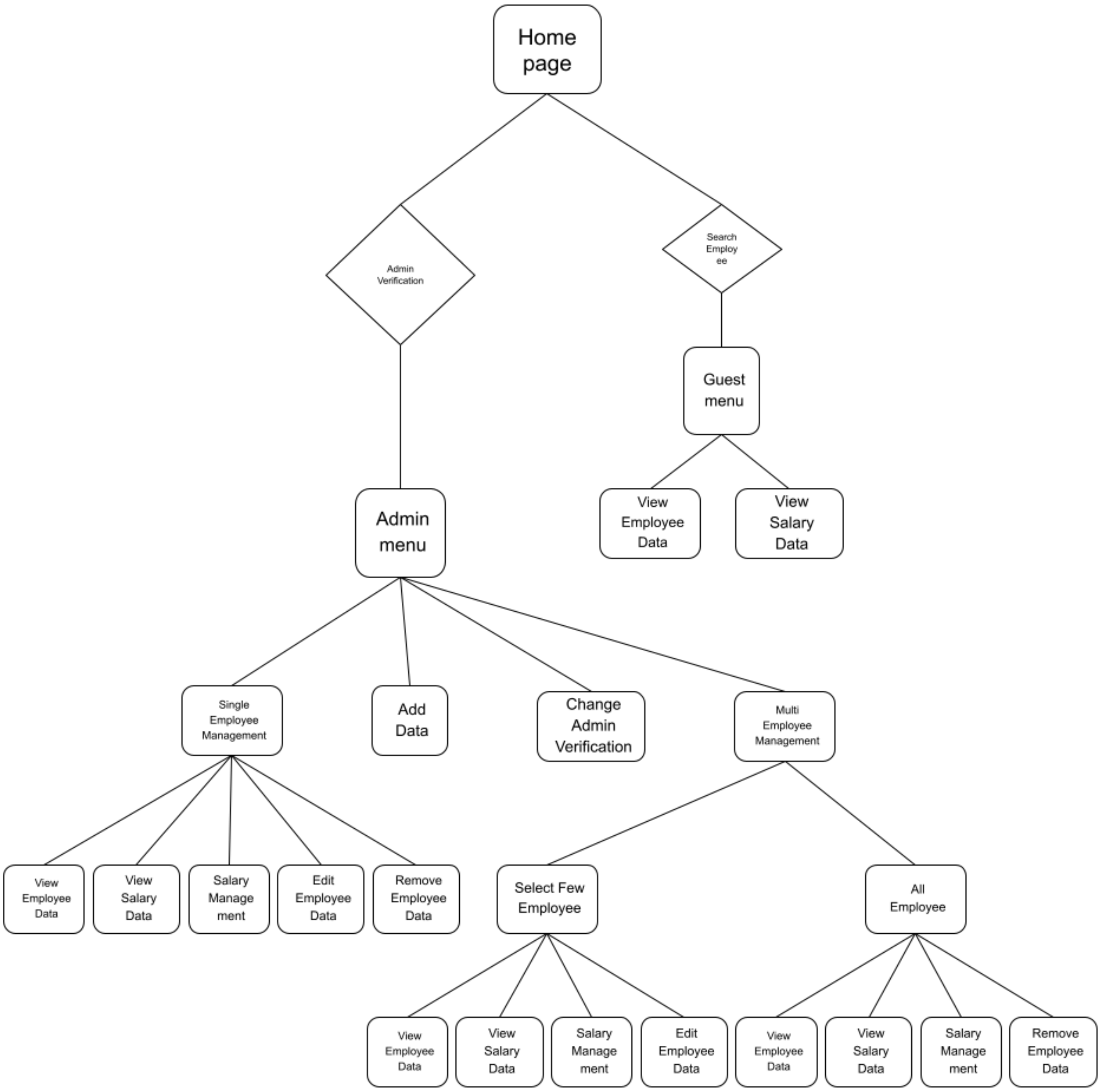

Figure 3.1: System Architecture Diagram

Chapter 4
IMPLEMENTATION

4.1. Interface/Front-End Design

This project employs a console-based interface, emphasizing simplicity and clarity for all users. The system leverages C++'s standard input and output functions to implement menus that guide users through various tasks. Interaction occurs via a text-based interface, presenting prompts and options for user engagement. All menus and submenus adhere to a consistent format, facilitating easy navigation. The developers incorporated thorough input validation, displaying error messages when users input invalid data, thereby maintaining system reliability. This design approach ensures a straightforward and efficient user experience for both administrators and regular users.[3]

4.2. Interface/Back-End Design

The back-end design of this Employee Management System is implemented in **C++**, utilizing file handling to store and manage employee data persistently. Text files are used as a simple database to save employee records, job roles, salary details, attendance logs, and any updates to employee information.

Core Functionalities:

1. **Employee Authentication :**system allows employees and admins to register, log in, and access their accounts. It verifies credentials (ID, password) from a stored file, distinguishing roles and ensuring secure access to sensitive information.
2. **Employee Details :** Admins can manage employee details such as name, contact, department, and position. These details are stored in a file, allowing easy updates and providing a history of changes for transparency and reporting purposes.
3. **Information Set(Data):** Text files are used to save employee data, such as personal and pay information. Effective reading, writing, and searching features guarantee seamless data management and accessibility even after the software has been shut down.
4. **Payment administration:** Administrators are able to oversee employee pay, including revisions. Every change in pay is documented in a file, giving future reference and audits a record of the adjustments.

4.3. Modules/Features

The Employee Management System is divided into several modules, each responsible for specific functionalities. Below are the key modules:

1. Authentication Module:

1. Handles user registration and login processes.
2. Differentiates between Admin and Employee roles.
3. Securely stores user credentials, such as username and password, in a file.

2. Employee Management Module:

1. Admins can add, update, and remove employee records (e.g., name, contact details, department, position).
2. Maintains a history of updates to employee records for transparency and auditing.

3. Stores employee data in a text file for persistence.

3. File Handling Module:

1. Implements functions to read, write, and update employee and salary data files.
2. Ensures data consistency and handles potential file access conflicts.
3. Manages data storage, ensuring information remains persistent across sessions.

4. Salary Management Module:

1. Admins can manage employee salaries, including updates, bonuses, and deductions.
2. Keeps track of salary changes and maintains a record of all adjustments.

5. User Interface Module:

1. Provides a text-based menu system for user interaction.
2. Ensures easy navigation and provides clear prompts and messages to guide users through actions.
3. Aims to simplify operations for both admins and employees with straightforward options.

Lists the modules/Features you have developed in this project.]

Chapter 5
USER MANUAL

5.1. System Requirement

5.1.1. Hardware Requirement

Device - A Laptop
Operating System - Microsoft Windows 11 pro
RAM- minimum 4 GB
Storage-minimum 1 GB free space

5.1.2. Software Requirement

The required software for the project are :
IDE - Codeblocks
Compiler-GNU GCC

5.2. User Interfaces

Here is the detailed list and title of the project module with interface picture or screenshot.

5.2.1. Panel A : Home page

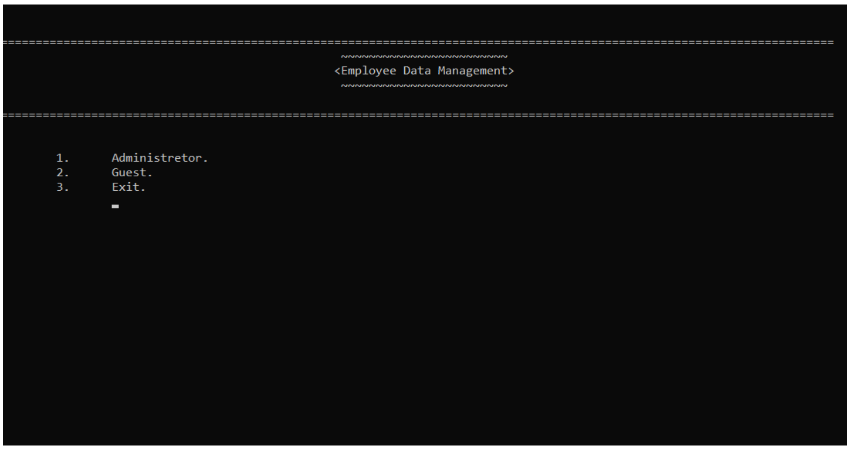

Figure 5.1: Home Interface

5.2.2. Panel B : Admin login interface (After pressing 1 in Panel A)

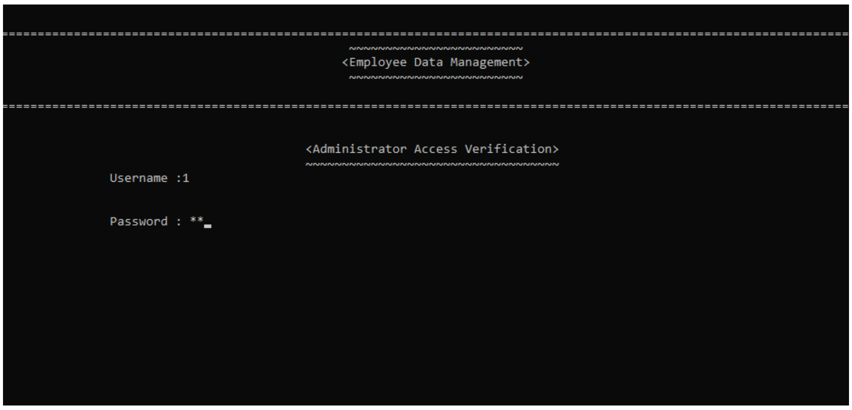

Figure 5.2: Admin Verification Interface

5.2.3. Panel C: Admin menu (After entering username and password in panel B)

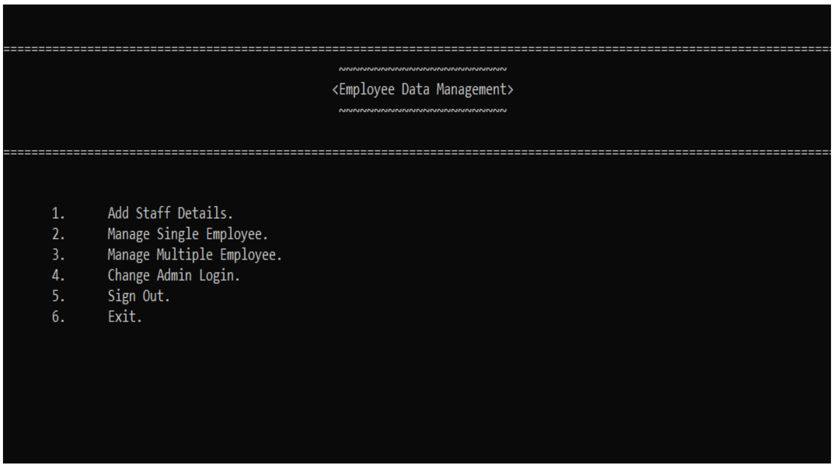

Figure 5.3: Admin Menu Interface

5.2.4. Panel D: Admin can add new staff details here(After pressing 1 in panel C)

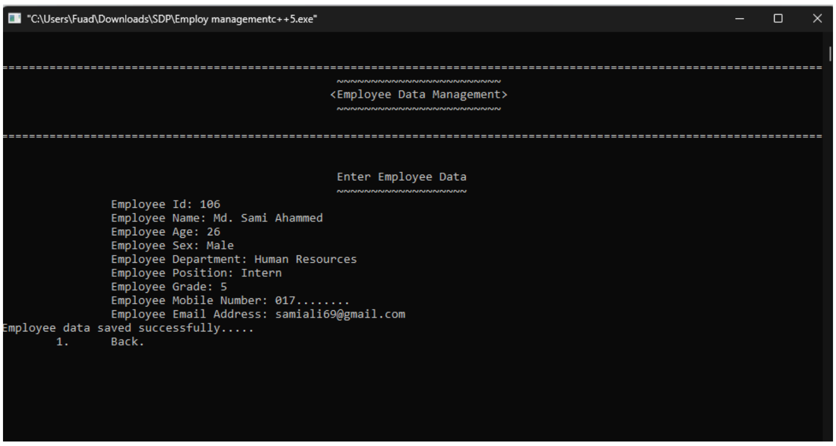

Figure 5.4: Employee Data Adding Interface

5.2.5. Panel E: Admin can search single employee details in this interface(After pressing 2 in panel C)

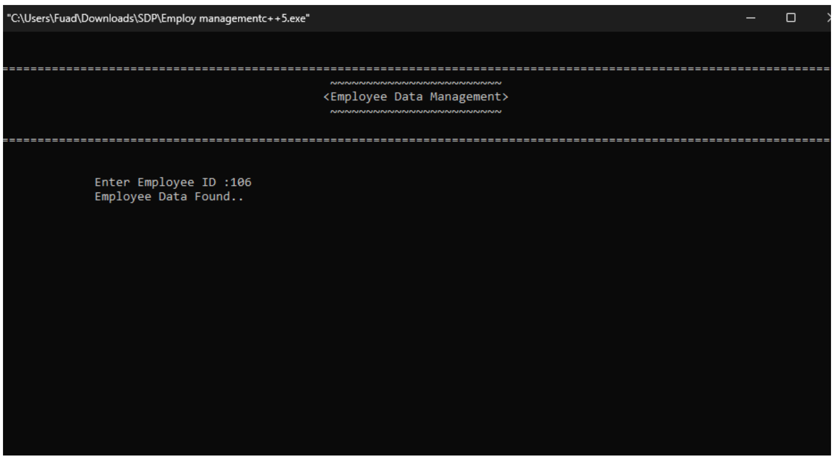

Figure 5.5: Employee Data Searching Interface

5.2.6. Panel F: Admin can handle single employee details in this interface(After entering employee id in panel E)

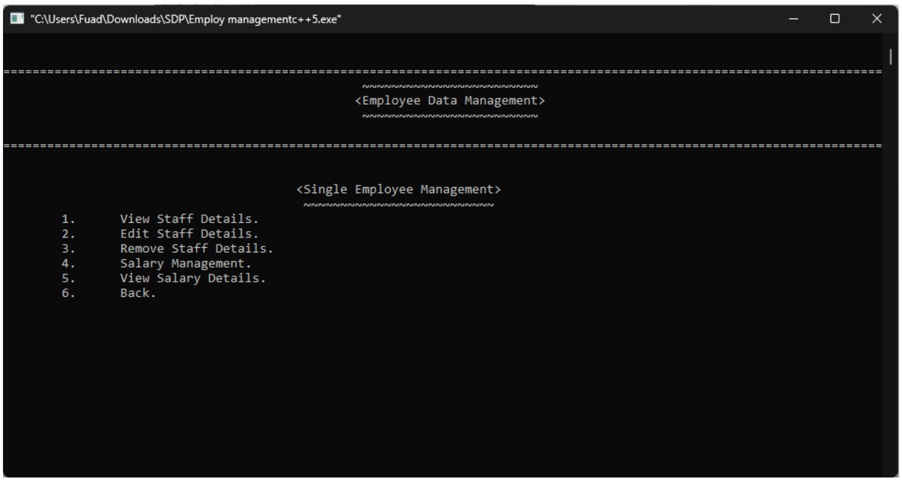

Figure 5.6: Single Employee Data Management Menu Interface

5.2.7. Panel G: Admin can view single employee details in this interface(After pressing 1 in panel F)

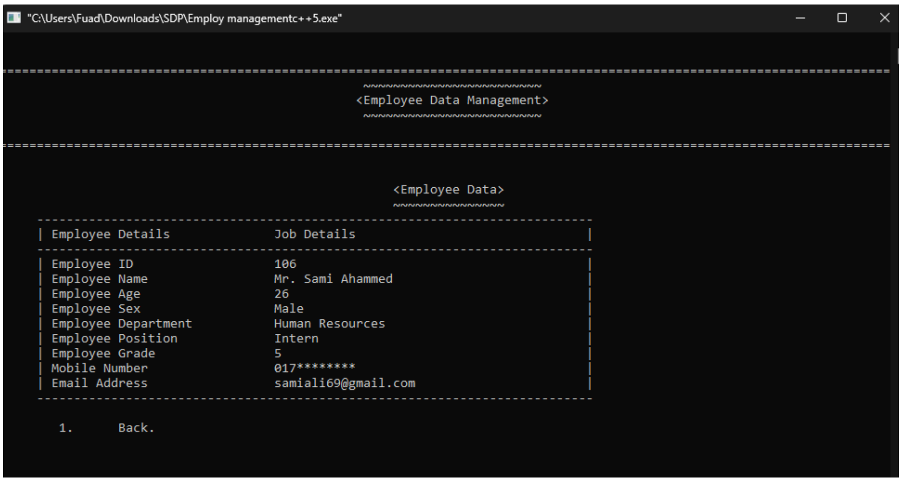

Figure 5.7: Single Employee Data Display Interface

5.2.8. Panel H: Admin can edit single employee details in this interface(After pressing 2 in panel F)

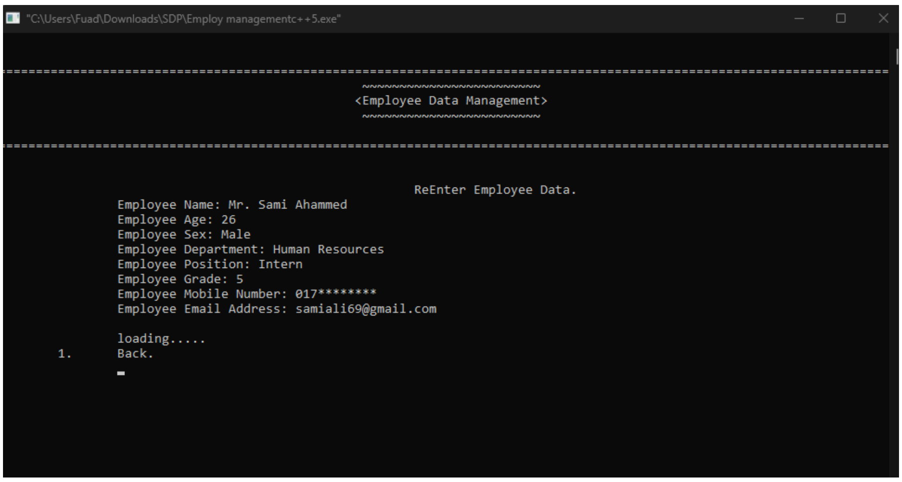

Figure 5.8: Single Employee Data Editing Interface

5.2.9. Panel I: Admin can remove single employee details in this interface(After pressing 3 in panel F)

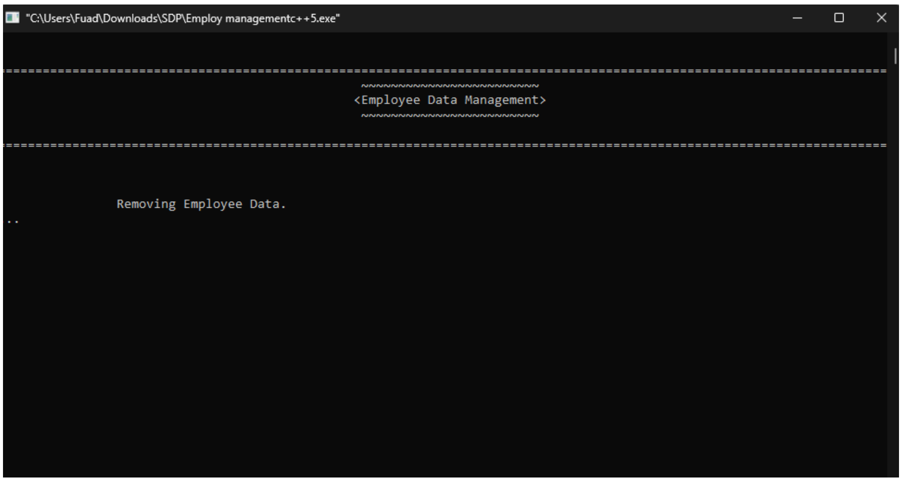

Figure 5.9: Single Employee Data Removing Interface

5.2.10. Panel J: Admin can handle single employee salary details in this interface (After pressing 4 in panel F)

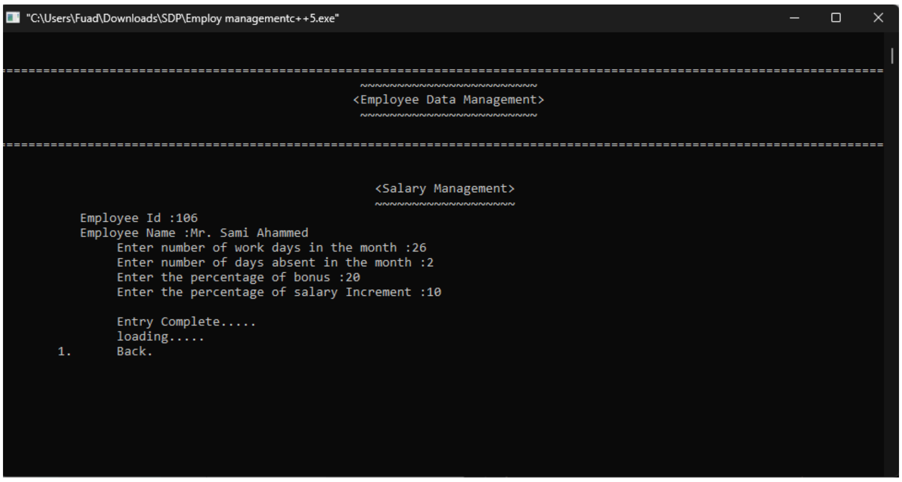

Figure 5.10: Single Employee's Salary Data Managing Interface

5.2.11. Panel K: Admin can view single employee salary details in this interface (After pressing 5 in panel F)

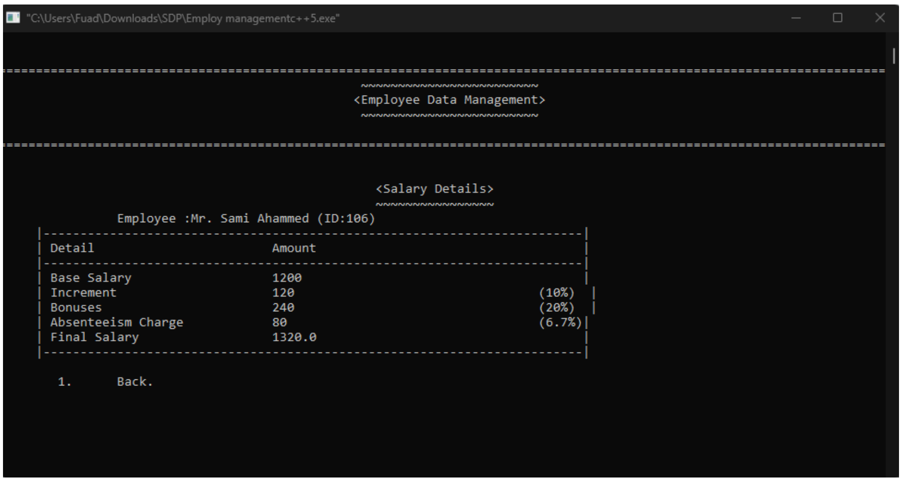

Figure 5.11: Single Employee's Salary Data Display Interface

5.2.12. Panel L: Admin can search for select few or all employees details in this interface(After pressing 3 in panel C)

Figure 5.12: Multiple Employees Data Selection Interface

5.2.13. Panel M: Admin can handle select few employees details in this interface (After entering a number less than 5 in panel L)

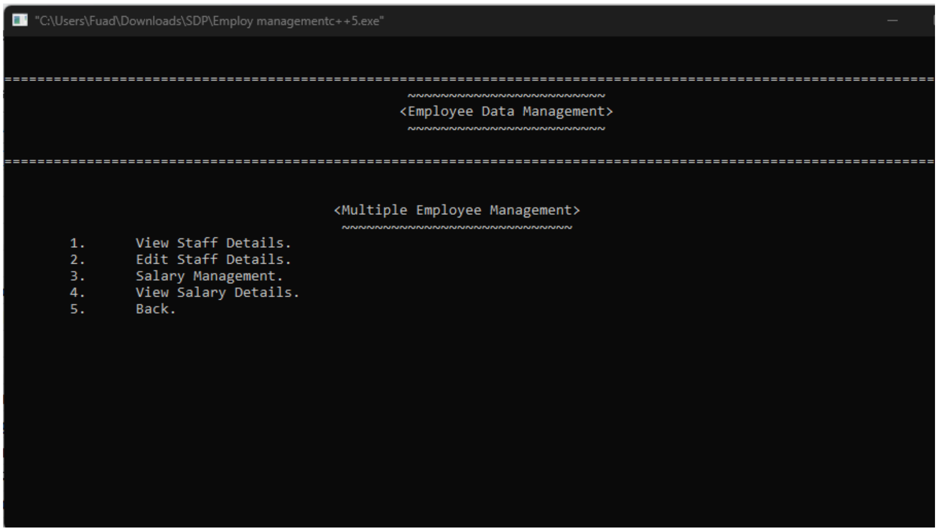

Figure :5.13: Multiple Employees Data Management Menu Interface 1

5.2.14. Panel N: Admin can view select few employees details in this interface (After pressing 1 in panel M)

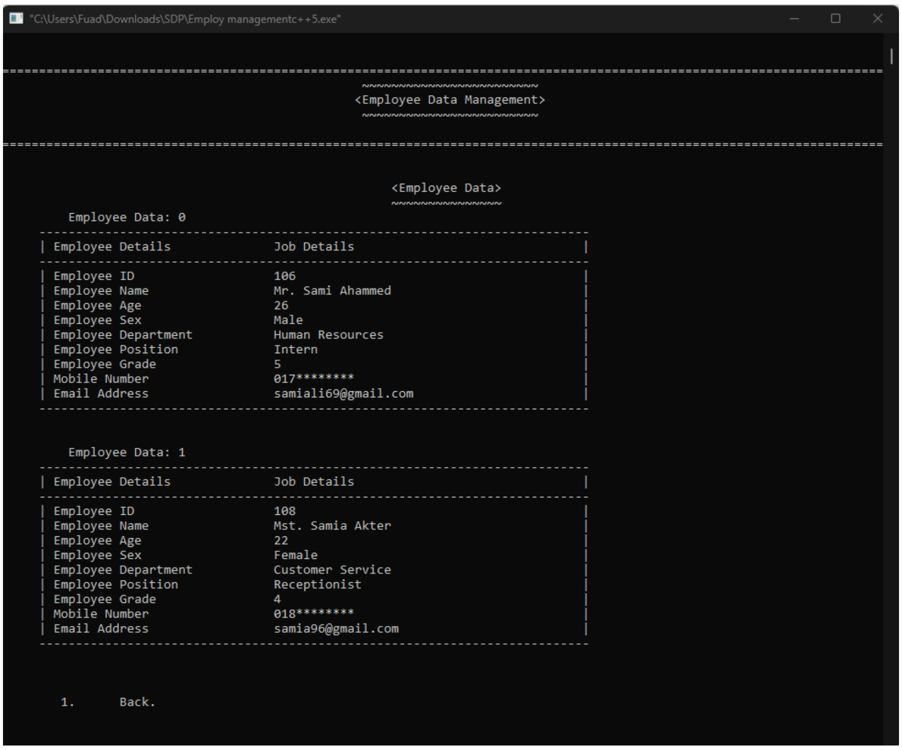

Figure 5.14: Multiple Employees Data Display Interface

5.2.15. Panel O: Admin can edit select few employees details(After pressing 2 in panel M)

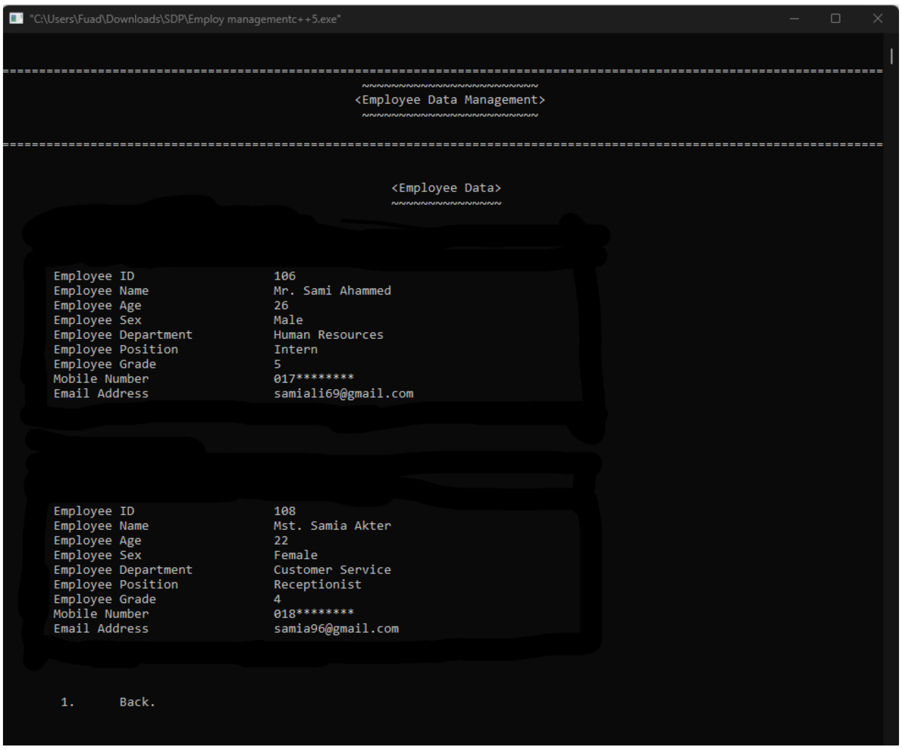

Figure 5.15: Multiple Employees Data Editing Interface

5.2.16. Panel P: Admin can handle select few employees salary details(After pressing 3 in panel M)

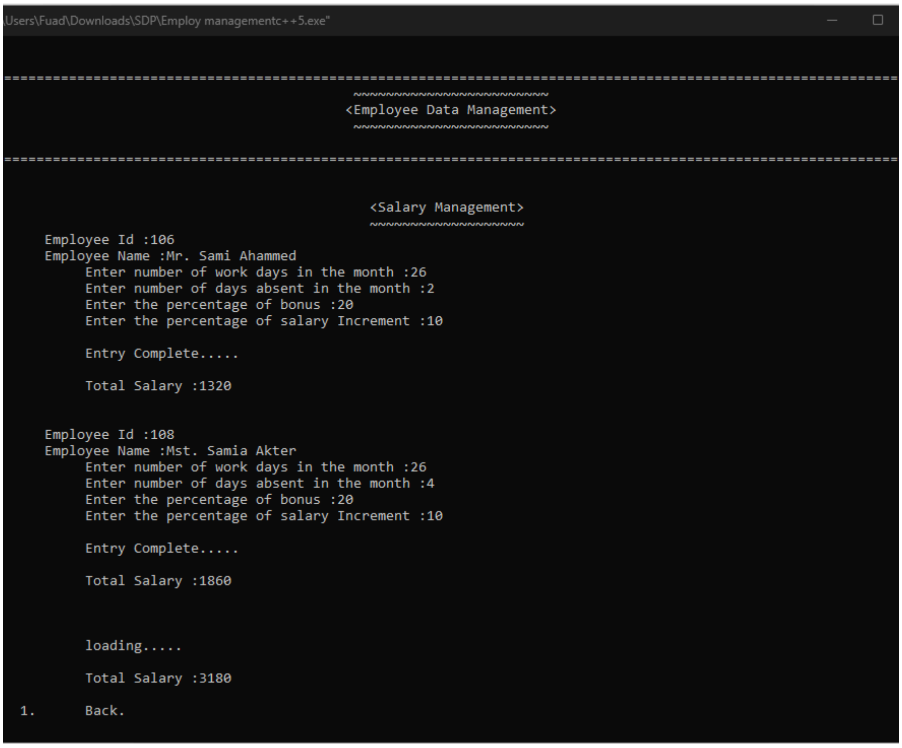

Figure 5.16: Multiple Employees' Salary Data Managing Interface

5.2.17. Panel Q: Admin can view select few employees salary details(After pressing 4 in panel M)

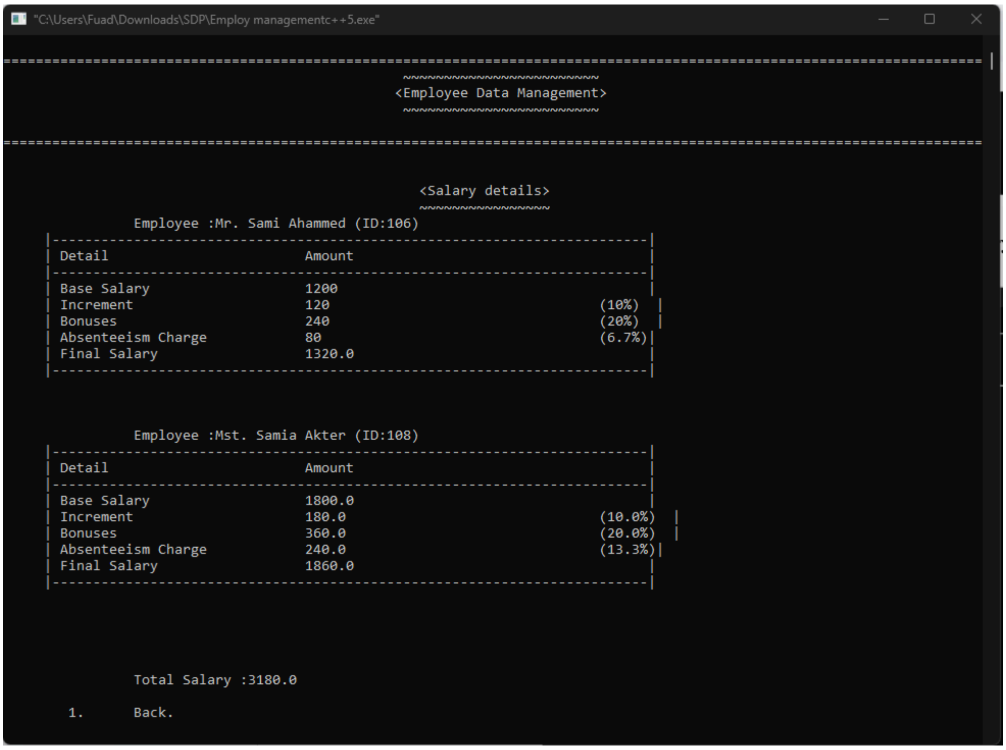

Figure 5.17: Multiple Employees' Salary Data Display Interface

5.2.18. Panel R: Admin can handle all employees details(After entering 0 in panel L)

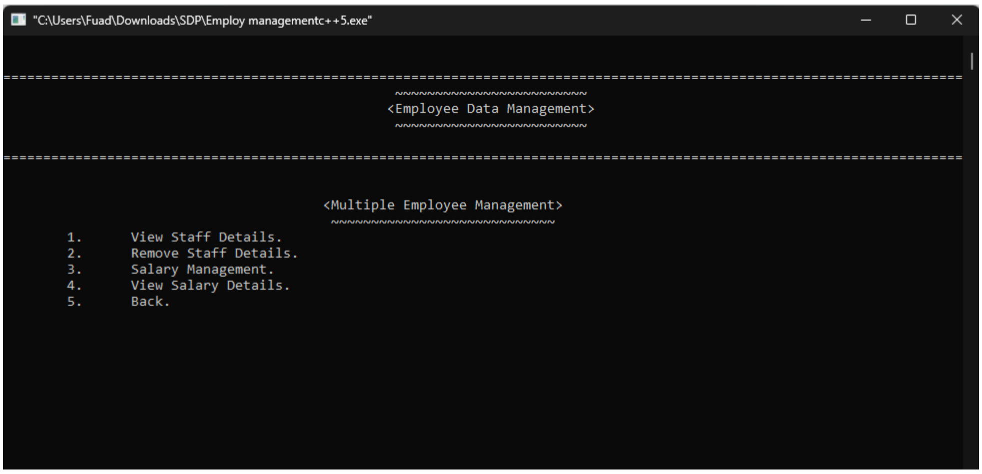

Figure 5.18: Multiple Employees Data Management Menu Interface 2

5.2.19. Panel S: Admin can change the username and password from this interface(After pressing 4 in panel C)

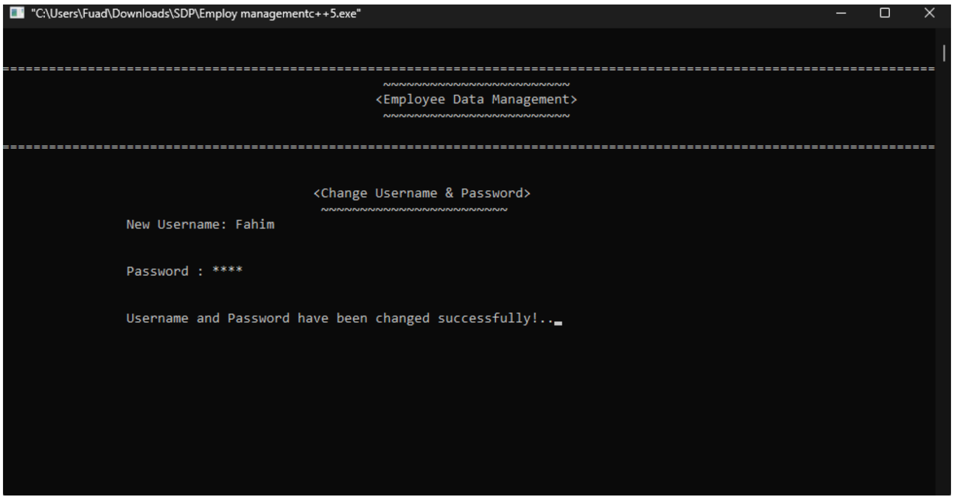

Figure 5.19: Admin Credential Updating Interface

5.2.20. Panel T: Guest login interface(press 2 in panel A).

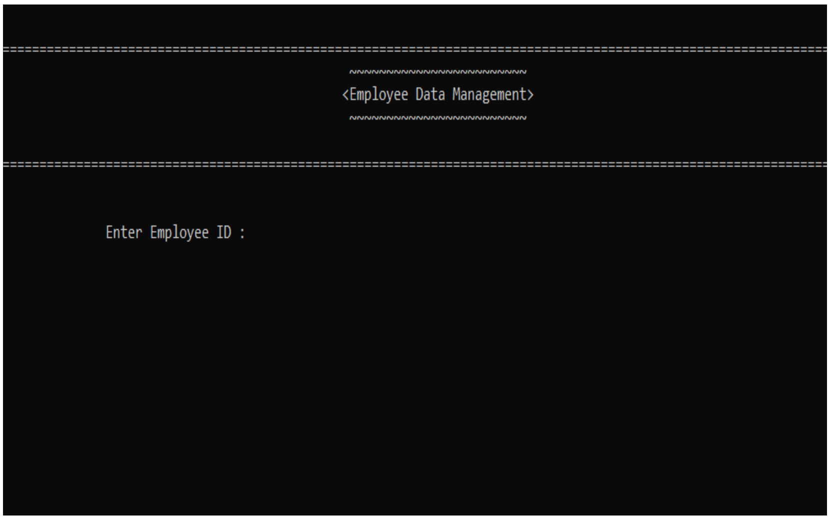

Figure 5.20: Guest Employee's ID Verification Interface

5.2.21. Panel U: Guest menu : Guests get two options where they can see their personal information and salary details.(After entering their ID in panel

T)

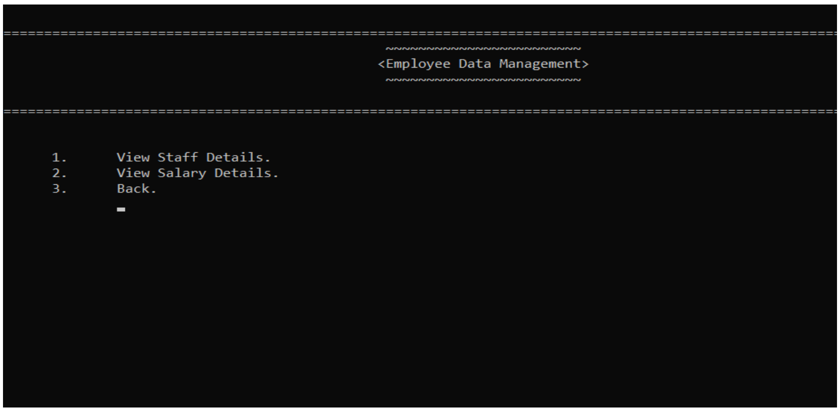

Figure 5.21: Guest Employee Menu Interface

5.2.22. Panel V: Guest can see their personal information(after pressing 1 in Panel U)

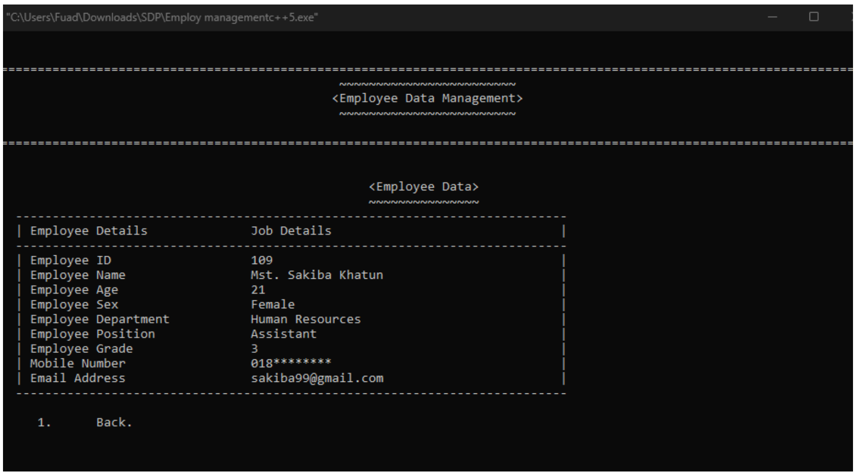

Figure 5.22: Guest Employee's Data Display Interface

5.2.23. Panel W: Guest an option to view their monthly salary(after pressing 2 in panel U)

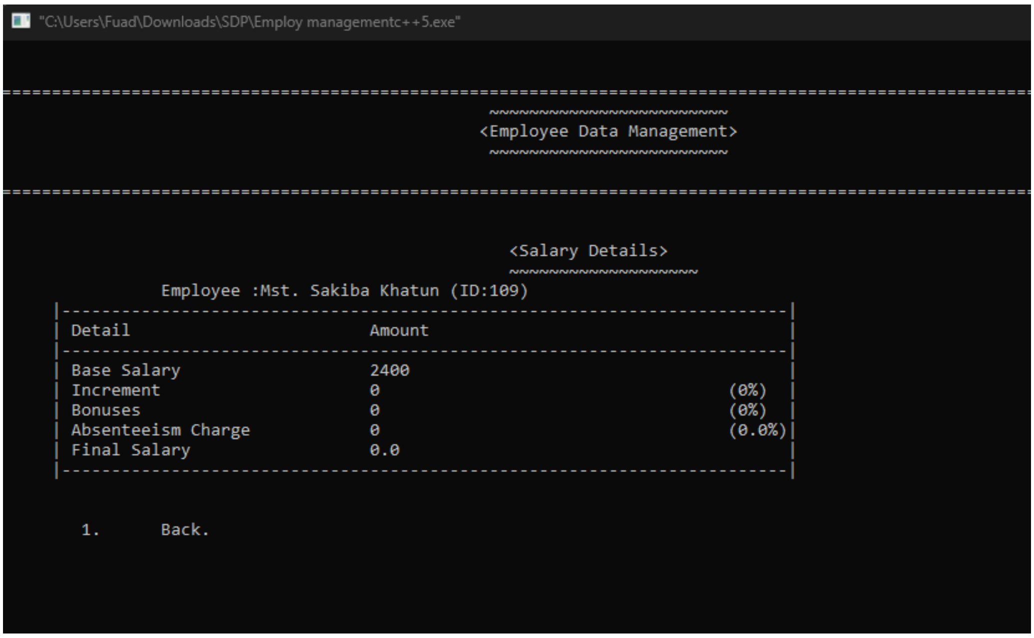

Figure 5.23: Guest Employee's Salary Data Display Interface

5.2.24: Login Credentials:

Admin Login Credentials:

- Username: Fahim
- Password: 1212

User Login Credentials:

- ID: 106,107,108,,,,,,,

Chapter 6
CONCLUSION

6.1. Conclusion

"Employee Management System" project has successfully created an easy-to-use and secure platform for managing employee information. The system helps improve work efficiency and supports the growth of the company with its simple features and smooth processes.The project not only addresses the current needs of the company but also sets the stage for future improvements.

6.2. Limitations

Here are some limitations of “Employee Management System” -

- **Limited Number of Employees :** The program struggles to handle a large number of employees and crashes when too much data is entered.
- **Slow Performance :** When there are too many employees record, the system takes more time to load or search data.
- **Limited Features :** The platform lacks advanced functionalities such as compensation processing, or employee evaluation monitoring, focusing instead on fundamental operations.
- **No Backup System :** If the system crashes there is no way to recover lost data, leading to potential data loss.
- **Failure to Compute Annual Salary :** The system is unable to calculate and record an employee's total yearly earnings (for all 12 months), which is a crucial feature for comprehensive payroll administration.

6.3. Future Works

As we finish the first phase of the "Employee Management System" project, there are many opportunities to improve and add new features to make the system even better for users. Here are some ideas for future updates:

- Multi- Platform Accessibility.
- Database Integration Setup.
- Use Advanced Software.
- Upgrade UI and UX .

References

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Appendix A